MODULE 09 · DOMAIN 1 - MONITORING, LOGGING, ANALYSIS, REMEDIATION, AND PERFORMANCE OPTIMIZATION

# Compute and EBS Performance

Instance families, burst credits, volume types and the ceiling that is actually limiting you.

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admin@sthithapragna.com

Storage performance has **three ceilings**: the volume, the instance, and the file system. The lowest one wins, and it is usually not the one you configured.

# Reading an instance type name

- **m7i.2xlarge**: family **m**, generation **7**, processor **i**, size **2xlarge**.
- Processor letters: **i** Intel, **a** AMD, **g** Graviton (AWS Arm). No letter means Intel on older generations.
- Extra letters: **n** higher network bandwidth, **d** local NVMe instance store, **e** extra memory or storage, **flex** lower cost for non-sustained load.
- A higher generation on the same family is almost always **cheaper per unit of work**, not just faster.

> ON THE EXAM
>
> Reading the name tells you most of what a question needs. m7i is general purpose, seventh generation, Intel. r7gd adds memory, Graviton and local NVMe.

# Families, and what each is for

| FAMILY | OPTIMISED FOR |
|---|---|
| T | Burstable. Low steady load with occasional spikes. Cheapest per hour |
| M | General purpose. Balanced CPU, memory and network. The default |
| C | Compute. High CPU per unit of memory: batch, encoding, HPC, web front ends |
| R, X, z | Memory. Databases, caches, in-memory analytics. X and z are extreme |
| I, D, H | Storage. Local NVMe or HDD, very high local IOPS: NoSQL, data warehouses |
| P, G, Inf, Trn | Accelerated. GPUs and AWS silicon for training and inference |

ON THE EXAM

Questions rarely ask which family. They describe a workload and expect you to know that "in-memory cache" means R, not M.

# Graviton, briefly

- AWS Arm-based processors. Generally **better price for the same performance** than comparable x86 instances.
- Well suited to **web servers, containers, microservices, caches and open-source databases**.
- It is a **different architecture**. Anything with compiled binaries or a licence tied to x86 needs checking first.
- Most managed services support it: RDS, ElastiCache, OpenSearch, Lambda, Fargate.

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>
> "Reduce cost without reducing performance" on a Linux workload with no exotic dependencies is very often a Graviton answer.

# How burstable instances work

- Each T instance has a **baseline CPU percentage**. Below it you **earn credits**; above it you **spend them**.
- One credit is **one vCPU running at 100 percent for one minute**.
- Baselines: **t3.micro 10%, t3.small and medium 20%, t3.large 30%, t3.xlarge and 2xlarge 40%.**
- **CPUCreditBalance** is the metric. When it reaches zero in standard mode, the instance is throttled to baseline.

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>
> A T instance is a bet that your average load is below baseline. If the average is above it, you are on the wrong family.

# Standard or unlimited mode?

STANDARD

## Throttle when credits run out

- Performance drops to baseline, hard
- Never costs more than the hourly price
- T2 default. Recommended for Spot
- Predictable bill, unpredictable performance

UNLIMITED

## Borrow, and maybe pay

- Bursts indefinitely using surplus credits
- Free if the 24-hour average is at or below baseline
- T3, T3a and T4g default
- Predictable performance, variable bill

THE TIEBREAKER

Unlimited averages over a rolling 24 hours. Sustained load above baseline is charged per vCPU-hour, and can cost more than a right-sized M instance.

# Diagnosing a burstable instance

- **Application slow, CPU pinned at exactly the baseline?** Standard mode with an exhausted credit balance. Look at CPUCreditBalance.
- **Bill higher than expected on T3?** Unlimited mode paying for surplus. Look at CPUSurplusCreditBalance.
- **The fix is usually the family, not the mode.** Consistent load above baseline belongs on M or C.
- **T3 gets no launch credits.** Only T2 standard receives them, at 30 per vCPU.

> ON THE EXAM
>
> A CPU graph flat-lining at 20 percent is not a coincidence. That is a t3.medium at its baseline, throttled.

# EBS volume types, current numbers

| TYPE | KEY FIGURES |
|---|---|
| gp3 | 3,000 IOPS and 125 MiB/s baseline at any size. Up to 80,000 IOPS, 2,000 MiB/s, 64 TiB |
| gp2 | 3 IOPS per GiB, bursts to 3,000. Max 16,000 IOPS at 5,334 GiB, 16 TiB |
| io1 | Up to 64,000 IOPS, 1,000 MiB/s. 50:1 IOPS per GiB |
| io2 Block Express | Up to 256,000 IOPS, 4,000 MiB/s, 64 TiB. 99.999% durability, sub-millisecond |
| st1 | Throughput HDD. Up to 500 MiB/s but only 500 IOPS. Sequential only |
| sc1 | Cold HDD. Up to 250 MiB/s, 250 IOPS. Cheapest per GiB |

ON THE EXAM

HDD volumes cannot be boot volumes. Pair st1 or sc1 with a small gp3 root volume.

# gp3 changed, and most study guides did not

- **Since September 2025:** gp3 supports up to **64 TiB**, **80,000 IOPS** and **2,000 MiB/s**.
- The widely published figures of **16 TiB, 16,000 IOPS and 1,000 MiB/s** are the old limits.
- The practical effect: workloads that used to require io1 or io2 now fit comfortably on **much cheaper gp3**.
- **gp3 has no burst.** What you provision is what you get, all the time. That is the point of it.

> ON THE EXAM
>
> If a question hinges on "gp3 cannot exceed 16,000 IOPS", that premise is out of date. Reach for io2 only above 80,000 IOPS or for the durability.

# The ratio constraint nobody mentions

- **IOPS per GiB is capped by volume type**, so a small volume cannot be given large IOPS however much you pay.
- **gp3 is 500:1.** A 20 GiB volume tops out at 10,000 IOPS. Reaching 80,000 needs at least **160 GiB**.
- **io1 is 50:1.** Reaching 64,000 IOPS needs **1,280 GiB**. **io2 Block Express** needs **256 GiB** for 256,000.
- This is a **hard limit at provisioning time**, not a performance target you might miss.

> ON THE EXAM
>
> "We need 30,000 IOPS on a 40 GiB volume" is not a tuning problem. It is impossible, and the volume must be bigger.

# gp2 or gp3?

GP2

## Performance from size

- 3 IOPS per GiB, so you buy capacity to get speed
- Bursts to 3,000 IOPS using a credit bucket
- BurstBalance metric, and a cliff when it empties
- Previous generation

GP3

## Performance you choose

- 3,000 IOPS and 125 MiB/s at any size, included
- Provision IOPS and throughput independently
- No burst and no credit cliff
- About 20 percent cheaper per GiB

THE TIEBREAKER

gp3 is cheaper and more predictable at almost every size. "Migrate gp2 to gp3" is a very common correct answer.

# gp3 or io2 Block Express?

GP3 WHEN

## Almost always

- Up to 80,000 IOPS and 2,000 MiB/s is enough
- 99.8 to 99.9 percent durability is acceptable
- Single-digit millisecond latency is fine
- Substantially cheaper

IO2 BLOCK EXPRESS WHEN

## The exceptions

- Above 80,000 IOPS, or above 2,000 MiB/s
- Sub-millisecond latency: under 500 microseconds typical
- 99.999 percent durability required
- Multi-Attach across up to 16 instances

THE TIEBREAKER

Three specific triggers: extreme IOPS, sub-millisecond latency, or five-nines durability. Absent all three, gp3 is the answer.

# When a hard disk is the right answer

- **st1 and sc1 are throughput devices.** Up to 500 and 250 MiB/s, but only 500 and 250 IOPS.
- Excellent for **large sequential reads and writes**: log processing, big data, Kafka segments, data warehouse scans.
- Terrible for anything random. Small scattered reads will crawl.
- **Neither can be a boot volume.** Pair them with a small gp3 root volume.

> ON THE EXAM
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> About half the price of gp3 per GiB. At petabyte scale on sequential workloads, that difference is the entire argument.

# The instance is often the real ceiling

- Every instance size has its own **EBS bandwidth and IOPS maximum**, independent of the volume attached.
- **Smaller sizes only burst to their headline number**, typically for about 30 minutes in 24 hours, then drop to roughly half.
- So a 4xlarge and an 8xlarge can quote the same maximum, and only the 8xlarge sustains it.
- Instance-level ceilings for gp3-attached storage reach about **260,000 IOPS and 12,500 MB/s** on the largest sizes.

> ON THE EXAM
>
> Provisioned 40,000 IOPS and seeing 20,000 after half an hour? That is the instance's EBS burst expiring, not the volume.

# Instance store, and Elastic Volumes

- **Instance store** is local NVMe. The highest IOPS available, and **lost on stop or terminate**.
- Right for **caches, scratch, buffers and replicated data**. Never for anything you need to keep.
- **Elastic Volumes** change an EBS volume's size, type, IOPS or throughput **with no downtime and no detach**.
- You must wait about **6 hours** before modifying the same volume again. Volumes can grow but never shrink.

> ON THE EXAM
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> "Change gp2 to gp3 without downtime" is Elastic Volumes. A snapshot and restore is the answer only if the volume must get smaller.

# EBS metrics that tell you what is wrong

- **VolumeQueueLength.** Requests waiting. Persistently high means the volume cannot keep up.
- **BurstBalance.** gp2, st1 and sc1 only. Trending to zero predicts the performance cliff before it arrives.
- **VolumeThroughputPercentage** and **VolumeConsumedReadWriteOps** on provisioned IOPS volumes show how much you are using.
- **gp3 has no BurstBalance metric**, because it does not burst. Its absence is not a fault.

> ON THE EXAM
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> Alarm on BurstBalance while you still have gp2 volumes. It is the only metric that warns you before the slowdown, rather than during it.

# Placement groups

| TYPE | WHAT IT DOES, AND WHEN |
|---|---|
| Cluster | Packs instances close together in one AZ. Lowest latency, highest throughput. HPC |
| Spread | Separate hardware, max 7 instances per AZ per group. Small critical sets |
| Partition | Groups on distinct racks, up to 7 partitions per AZ. HDFS, Cassandra, Kafka |

ON THE EXAM

Cluster maximises performance and concentrates risk. Spread maximises isolation. Partition is the compromise for distributed data stores.

# Network performance and right-sizing

- **ENA** is enhanced networking, standard on current instances. **EFA** adds OS-bypass for tightly coupled HPC and machine learning.
- **Compute Optimizer** analyses CloudWatch metrics and recommends instance types, EBS volumes, Auto Scaling groups and Lambda memory.
- It cannot see memory unless the **CloudWatch agent** is publishing it. Without that, recommendations are CPU and network only.
- Default lookback is **14 days**. Enhanced infrastructure metrics extends this to **3 months**, for a charge.

> ON THE EXAM
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> Compute Optimizer and module one's memory lesson connect here. No agent means no memory metric, which means no memory-aware recommendation.

> A reporting application on a t3.medium runs a nightly job. For the first 40 minutes it performs well, then slows dramatically and takes four more hours. CloudWatch shows CPUUtilization flat at exactly 20 percent for the rest of the run.
>
> The instance is in standard credit mode. During the day the application is almost idle.

## What is happening, and what is the best fix?

A. The EBS volume's BurstBalance is exhausted. Move to a larger gp2 volume

B. CPU credits are exhausted and the instance is throttled to baseline. Move the job to a compute-optimised instance

C. Switch the instance to unlimited mode and leave everything else unchanged

D. The instance's EBS burst has expired. Attach a second volume and stripe across both

◆ PAUSE AND DECIDE

## What is happening, and what is the best fix?

A EBS BurstBalance exhausted

*Would show as high VolumeQueueLength and slow I/O, not as CPU pinned at exactly the baseline figure.*

B CPU credits exhausted; change family

20 percent is precisely t3.medium's baseline. A four-hour job above baseline is not a burstable workload.

C Switch to unlimited mode

*The most tempting answer. It fixes the throttling and bills surplus credits for four hours every night, potentially above the cost of a right-sized instance.*

D Instance EBS burst expired

*Real behaviour, wrong symptom. That limits storage throughput and would not hold CPU flat at exactly 20 percent.*

THE SIGNAL

"Flat at exactly 20 percent" is the giveaway. That is not a workload, it is a ceiling, and it is t3.medium's baseline.

> A database team requires 30,000 IOPS and sub-millisecond latency for a 40 GiB transactional volume. An engineer creates a gp3 volume of 40 GiB and attempts to provision 30,000 IOPS. The request is rejected.
>
> The team asks whether this is a quota that can be raised.

**What should the engineer explain and recommend?**

A. It is a soft quota. Request an increase for gp3 provisioned IOPS

B. gp3 is capped at 500 IOPS per GiB, so 40 GiB allows 20,000. Increase the volume size, and use io2 Block Express for sub-millisecond latency

C. gp3 cannot exceed 16,000 IOPS. Use io1 instead

D. Attach four 40 GiB gp3 volumes and stripe them with RAID 0

◆ PAUSE AND DECIDE

## What should the engineer explain and recommend?

A. Request a quota increase

*The IOPS-per-GiB ratio is an architectural limit of the volume type, not an adjustable account quota. Support cannot raise it.*

B. Ratio limit; resize, and io2 for latency

Correct on both halves. 500:1 caps a 40 GiB gp3 at 20,000, and sub-millisecond latency is an io2 Block Express characteristic, not a gp3 one.

C. gp3 cannot exceed 16,000

*Out of date since September 2025. gp3 now reaches 80,000 IOPS, so the stated reason is wrong even though io1 would work.*

D. RAID 0 across four volumes

*Would deliver the IOPS, and adds four failure domains, complexity and no latency guarantee. It also ignores why the request failed.*

THE SIGNAL

Two requirements, and the latency one is decisive. gp3 is single-digit milliseconds; sub-millisecond is an io2 Block Express property.

> An analytics workload on an m6i.4xlarge reads large sequential files. Its 4 TiB gp3 volume is provisioned at 40,000 IOPS and 1,000 MiB/s. Throughput reaches about 1,000 MiB/s for the first half hour of each run, then settles at roughly half that for the remaining three hours.
>
> The volume's BurstBalance metric shows no data, and VolumeQueueLength is moderate throughout.

## What is the most likely cause?

A. The gp3 volume's burst credits are exhausted

B. The instance's EBS bandwidth burst has expired, dropping it to its sustained baseline

C. The provisioned throughput setting is being throttled by the 500:1 IOPS ratio

D. The file system needs a larger block size

◆ PAUSE AND DECIDE

## What is the most likely cause?

A **gp3 burst credits exhausted**

*gp3 has no burst mechanism, which is exactly why BurstBalance shows no data. The absence of that metric disproves this option.*

B **Instance EBS burst expired**

Smaller instance sizes hold their maximum EBS bandwidth for about 30 minutes in 24 hours, then drop to roughly half. The timing and the ratio both match.

C **The 500:1 ratio**

*That ratio limits IOPS relative to capacity at provisioning time. A 4 TiB volume is nowhere near constrained, and it does not cause a mid-run drop.*

D **Larger block size**

*Might improve efficiency generally, but would not produce a clean halving exactly 30 minutes into every run.*

THE SIGNAL

Half an hour, then half the throughput. Both numbers point at the instance, and the missing BurstBalance metric rules out the volume.

A finance team must reduce EC2 and EBS spend without reducing performance. The estate is 400 Linux instances running containerised Java microservices, plus 900 gp2 volumes, most of them under 200 GiB. There are no commercial software licences involved.

Compute Optimizer reports that most instances are over-provisioned on CPU, but reports no memory findings at all.

## Which combination should the team implement?

A. Migrate the gp2 volumes to gp3, move suitable workloads to Graviton instances, and install the CloudWatch agent so memory recommendations become available

B. Migrate all volumes to io2 Block Express and buy Reserved Instances for the current sizes

C. Move all instances to T3 in unlimited mode and enlarge every gp2 volume

D. Enable enhanced infrastructure metrics and wait three months before acting

◆ PAUSE AND DECIDE

## Which combination should the team implement?

**A** gp3, Graviton, and the agent

gp3 is cheaper and more predictable, containerised Linux Java is ideal for Graviton with no licence blockers, and the agent explains the missing memory findings.

**B** io2 Block Express plus RIs

*io2 is far more expensive than gp2, so this increases storage cost. Reserving over-provisioned sizes locks in the waste for years.*

**C** T3 unlimited everywhere

*Microservices under steady load run above baseline continuously, so surplus credit charges apply constantly. Enlarging gp2 buys IOPS the expensive way.*

**D** Wait three months

*Delays every saving to improve data that is already conclusive on CPU. It is also the only option that changes nothing.*

THE SIGNAL

"No memory findings at all" is not a Compute Optimizer failure. It is module one's lesson: no agent, no memory metric, nothing to analyse.

# The five things to carry forward

- **Three ceilings: volume, instance, file system.** The instance is the one people forget, and it bursts for about 30 minutes.
- **gp3 now reaches 64 TiB, 80,000 IOPS and 2,000 MiB/s.** The old 16,000 figure is stale.
- **IOPS per GiB is a hard ratio.** A small volume cannot be given large IOPS at any price.
- **CPU flat at exactly the baseline means credits are gone.** The fix is usually the family, not the mode.
- **No CloudWatch agent means no memory recommendations** from Compute Optimizer.

> ON THE EXAM
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> Next: storage and database performance - S3, EFS, FSx, S3 Files, RDS, DynamoDB and ElastiCache.